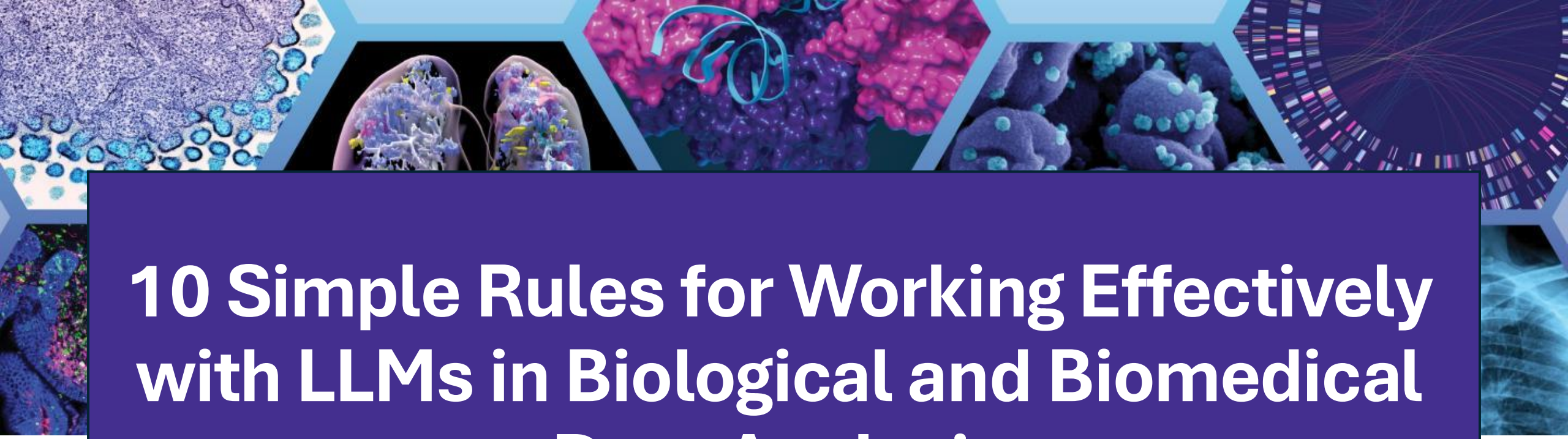


10 Simple Rules for Working Effectively with LLMs in Biological and Biomedical Data Analysis

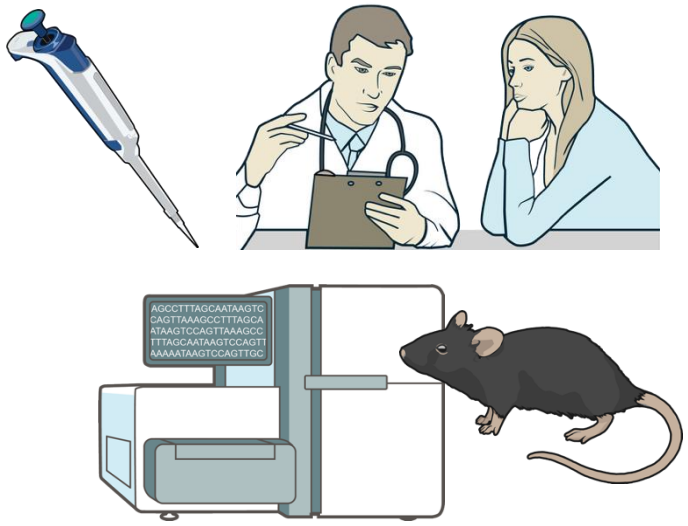
Joanne Berghout, PhD
Human Computational Genomics Specialist
Bioinformatics and Computational Biology Branch (BCBB),
OCICB, NIAID, NIH
17 June 2026

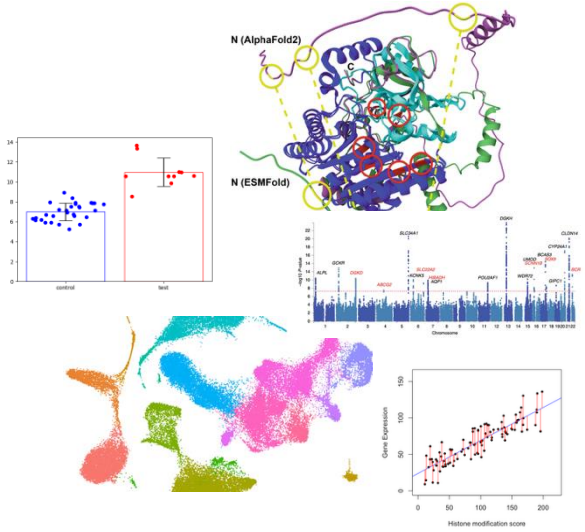


Gemini, visit https://go.hhs.gov/gemini
Claude, visit https://go.hhs.gov/claudeai
ChatGPT, visit https://go.hhs.gov/chatgpt

The Biomedical Research Process:

Whether you use an LLM or not, is the **data** that gets you to the knowledge and discovery









ChatGPT Claude Gemini

Stepping into the gap and “vibe coding”

- Code generation is no longer the bottleneck
  - Code *is* a language
- Doing computational biology is about more than making code that *runs* or working your way to a p-value that *exists*
- If you don’t make the decisions, the LLM will.
  - The LLM will never fail or refuse to create an output… but it will silently make a choice for you.
  - Typically, the choice it makes is that anything unsaid doesn’t matter

10 Simple Rules:

A. Document your design

1. Have a clear hypothesis
2. Describe your design and metadata explicitly
3. Have a validation plan

B. Know your data

4. Look at your data
5. Plot distributions, pay attention to missing values
6. Make your data rectangular and unambiguous
7. Don’t overwrite your raw data

C. Take accountability

8. Be accountable for all analysis design choices. Align with literature or be ready to defend differences
9. Sanity check outputs
10. Archive a reproducible end-to-end version of your analysis code

Document your design

#1: Have a clear hypothesis

- Specific
- Measurable
- Testable
- Falsifiable
- Relevant
- … Simple doesn't hurt

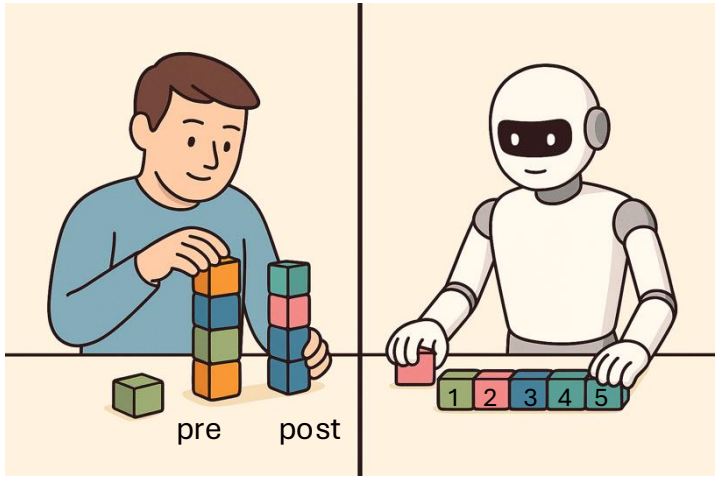
- **Bad:** Can we find a biomarker in this proteomics data?
- **Better:** Are there any individual proteins where abundance at d30 is associated with improved progression-free cancer survival at 24 months?

- **Bad:** We have 16S data from 35 people on different diets, what does their microbiome look like?
- **Better:** Independent of total calories, do participants on a high fiber diet have a greater microbiome diversity index than patients who eat more processed foods?

- **Bad:** Does this rare disease patient have variants in metabolic genes?
- **Better:** Does this rare disease patient have any variants that meet ACMG criteria for pathogenic in genes associated with monogenic metabolic disorders?

#2: Describe your experimental design and metadata explicitly





- Your experimental design is invisible and will be ignored. This can cause silent critical mismatch between an LLM’s analysis and your intent.
- An LLM will not infer anything from your raw data sheet
  - It does not read your words as words and does not know about your batches, replicates, relationships, or dependencies.
  - It does not know about your cohort or assay details
- Important data about your data (age, sex, genetic ancestry, clinical covariates, mouse strain, parent-child relationships, tissue types, etc) must be explicit. Metadata includes units, references, and versions.

#3. Have a validation plan

- This connects directly to rule #1 (hypothesis)
- Mitigates risk of data-dredging errors
  - p-hacking
  - Ignoring multiple testing
  - Inappropriate conclusions
- Iterating with an LLM is really, really easy and humans are really, really good at plausible model invention
- It's also just good science



Know your data

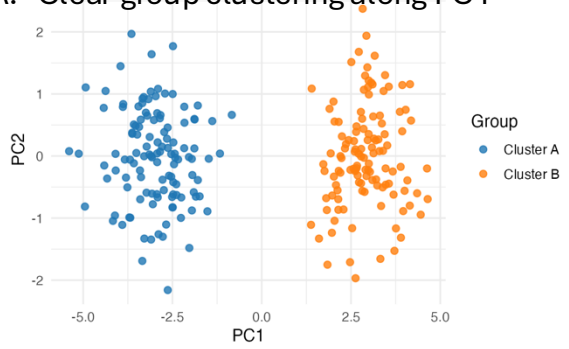
#4. Look at your data

Ensure counts and ranges meet your expectations & seem reasonable

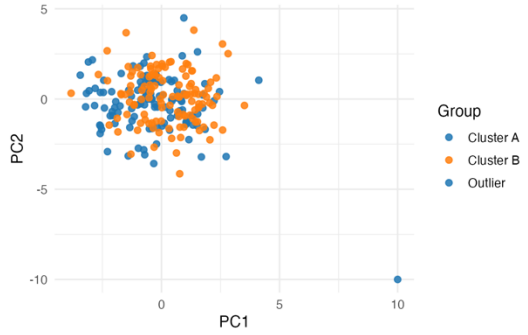
	Case	Control
Count	400	600
Sex (M)	233	288
Age (years)	46.2 (21-78)	44.9 (38-87)
Ancestry		
EUR	260	390
AFR	72	108
EAS	44	66
SAS	4	6
AMR	20	30
Heart attack	56	66
Diabetes	52	77
Statin Rx	184	192

Locate clusters or structured variance in your data

A. Clear group clustering along PC1



B. Cluster overlap along PC1, outlier

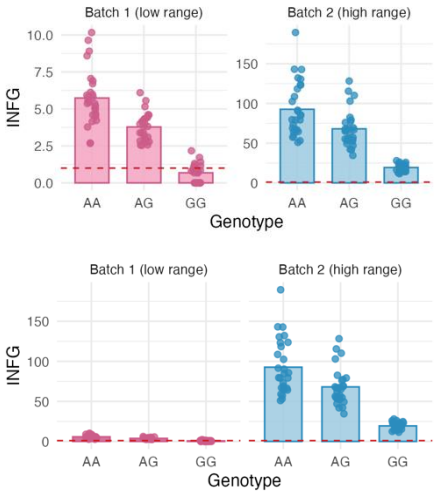


Before you begin an analysis, ensure your data looks “right”. If there are unexpected values or problems – you need to catch them here.

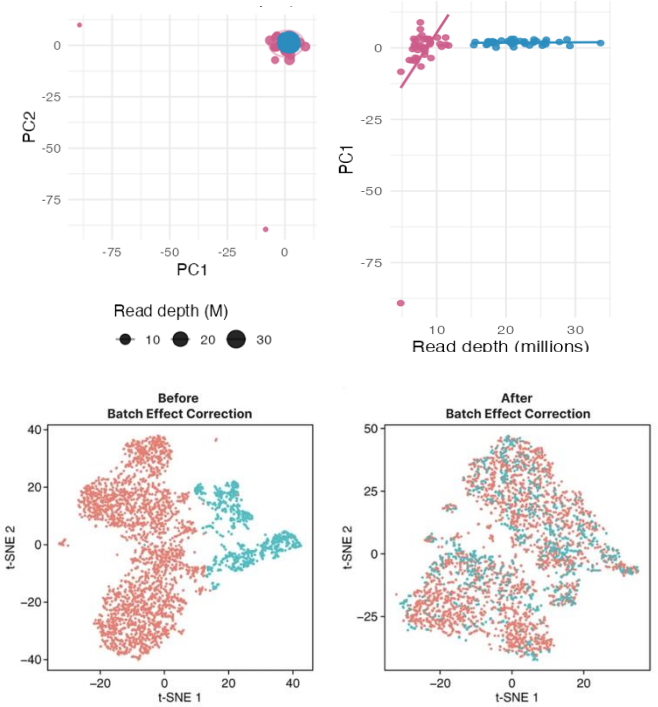
Plus you can learn a lot.

#4. Look at your data

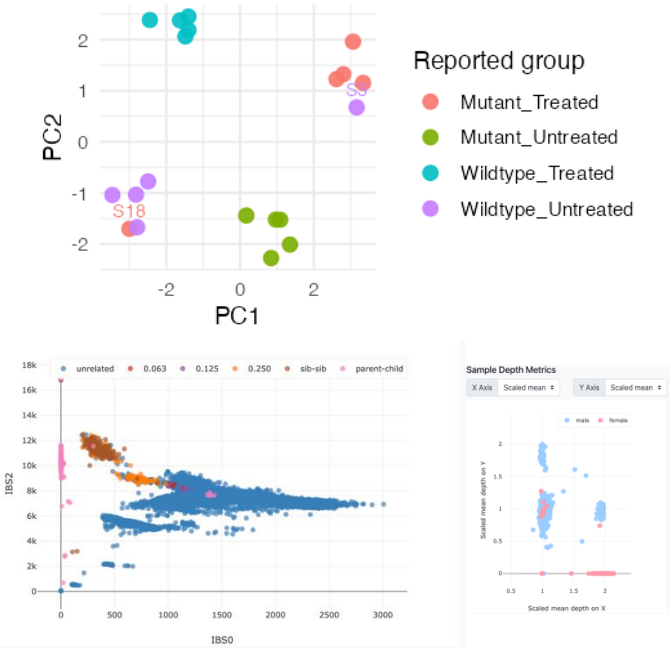
Batch effects in different ranges



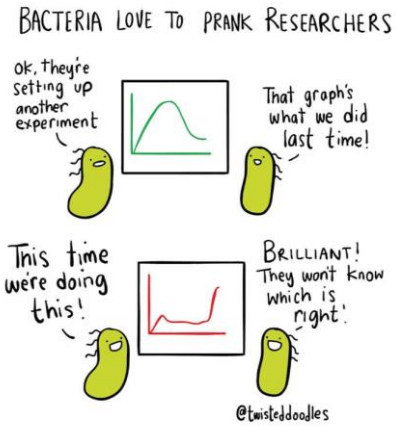
Distinguish biological versus technical variance (and prepare to correct technical)



QC for label swaps or incomplete documentation

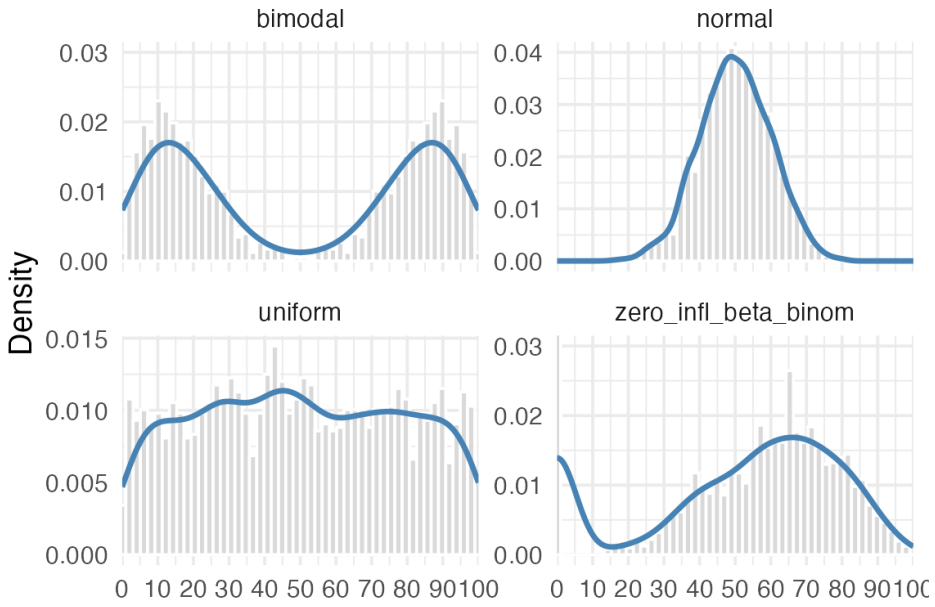


Look for weird stuff



#5. Plot distributions, pay attention to missing values

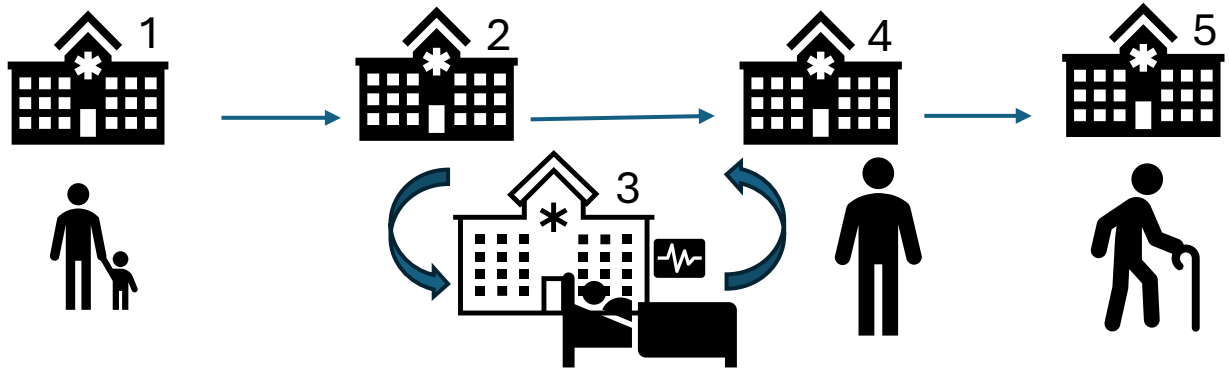
Same mean, different distributions



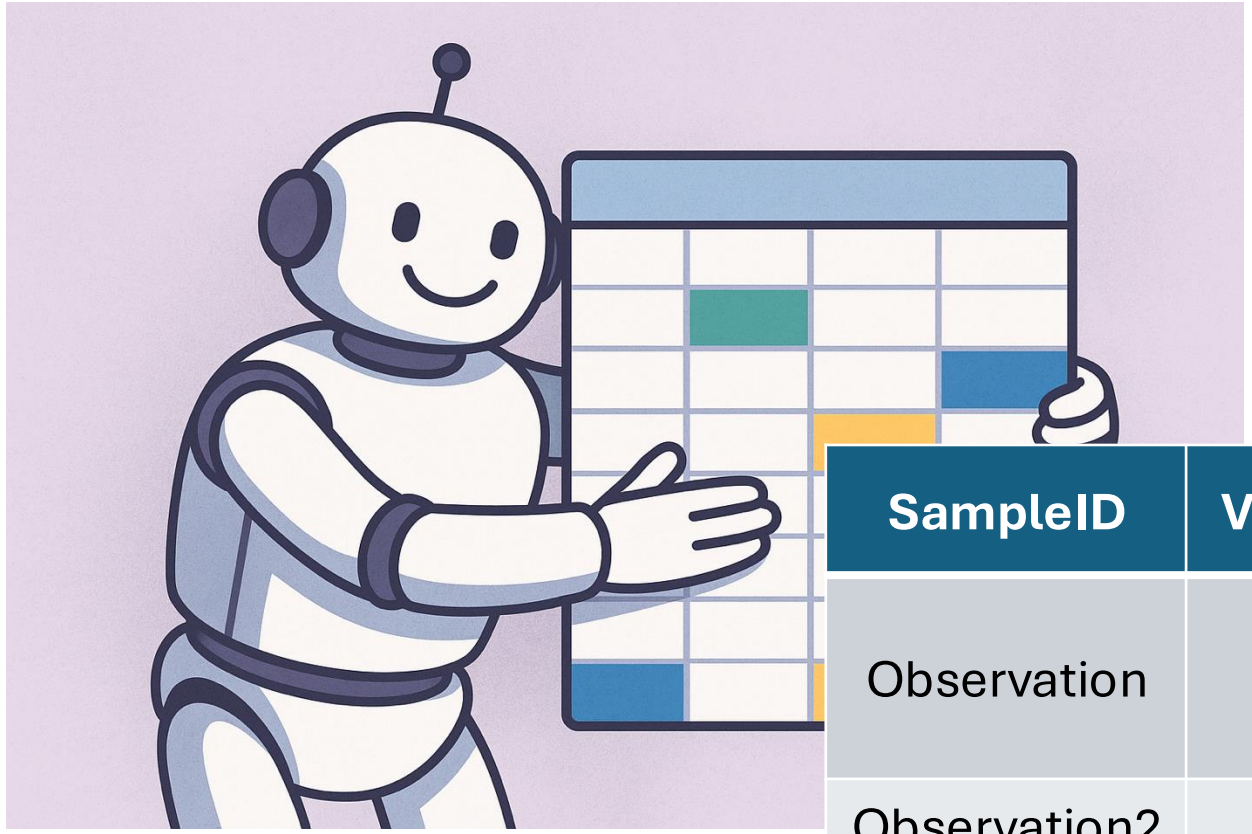
Zeroes and missingness are complicated (and not synonyms):

- Dataset completeness
- 'True absence' vs 'not documented'
- Observation missing not at random
- Source data fragmentation or differences in annotation
- Can be HIGHLY influential on statistics and conclusions

Fragmentation of Electronic Health Records:



#6. Make your data rectangular and unambiguous



SampleID	Variable	Variable2	Variable3
Observation	Value	Value	Value
Observation2 …	…	…	…

https://cran.r-project.org/web/packages/tidyr/vignettes/tidy-data.html
https://data.europa.eu/apps/data-visualisation-guide/intro-to-tidy-data

#6. Make your data rectangular and unambiguous

- Analysis-ready data is tabular and well-organized
- Tidy data is easy to summarize, transpose, join, pivot, aggregate, filter, transform, subset, apply operations, and visualize
- Each variable (column) describes only one thing
  - Nuance can be spread over multiple columns
  - Each covariate, measurement, annotation, etc has it's own variable
  - Each variable is only one 'type' (e.g. numeric, character, date, logical)
- Each intersection has a single value recorded per observation
  - Text or categorical values use consistent, mutually exclusive terms that reflect your study intent (ontologies can be very useful: HPO, EFO, GO, SNOMED)
  - Double check your 0, NA
- Avoid use of common delimiters unless intended as delimiters (commas, tabs, line-breaks, etc)

#6. Make your data rectangular and unambiguous

Missing column name

spaces

Inconsistent

Participant		Phenotypes	Gene or genes	Date
P0001112	45F	cough, pulmonary tuberculosis, meningitis, recurrent infection, bronchiectasis, HIV infection	*IFNgR-1*	06/21/2008
P0001113	31M	lower respiratory tract infection, Coughing	*CFTRdel508*	Jan 1, 2022
P0001114	8yo female	affected?	*AIRE*	2012

Non-standard

Two pieces of information

Two pieces of information

Nonstandard

inconsistent

Ambiguous

Participant	Age	Sex	Affected	Phenotypes	HIV	Gene	GRCh38_coord	Enrollment_Date_yyyymmdd
P0001112	45	F	Yes	cough, pulmonary tuberculosis, meningitis, recurrent infection, bronchiectasis, HIV infection	TRUE	IFNGR1	6-137204354-TA-T	2008-Jun-21
P0001113	31	M	Yes	lower respiratory tract infection, cough	FALSE	CFTR	7-117559590-ATCT-A	2022-Jan-01
P0001114	8	F	Unknown	*NA*	FALSE	AIRE	21-44289773-C-T	2012-Feb-04

#6. Make your data rectangular and unambiguous

Value not recognizable as numeric may be turned into a NA

Sample	day1 tnf (ng/ul)	day1 tgfb (ng/ul)	day5 tnfa (ng/ul)	day5 tgfb (ng/ul)
WT-1	13 ng/ul	3	450	144
WT-2	18	7	237, 272	200
MUT-1	21	11	41	95
MUT-2	9	8	dead	

Two pieces of information

Punctuation in values, headers, or merging cells may break rectangular alignment

Sample	Geno	Rep	Time	Cytokine	Value
WT-1	WT	1	1	TNF	13
WT-1	WT	1	1	TGFB	3
WT-1	WT	1	5	TNF	450
WT-1	WT	1	5	TGFB	144
WT-1	WT	2	1	TNF	18
WT-1	WT	2	1	TGFB	7
WT-1	WT	2	5	TNF	237
WT-1	WT	2	5	TNF	272
WT-1	WT	2	5	TGFB	200
…	…	…	…	…	…
MUT-2	MUT	2	1	TNF	9
MUT-2	MUT	2	1	TGFB	8
MUT-2	MUT	2	5	TNF	
MUT-2	MUT	2	5	TGFB	

#6. Make your data rectangular and unambiguous

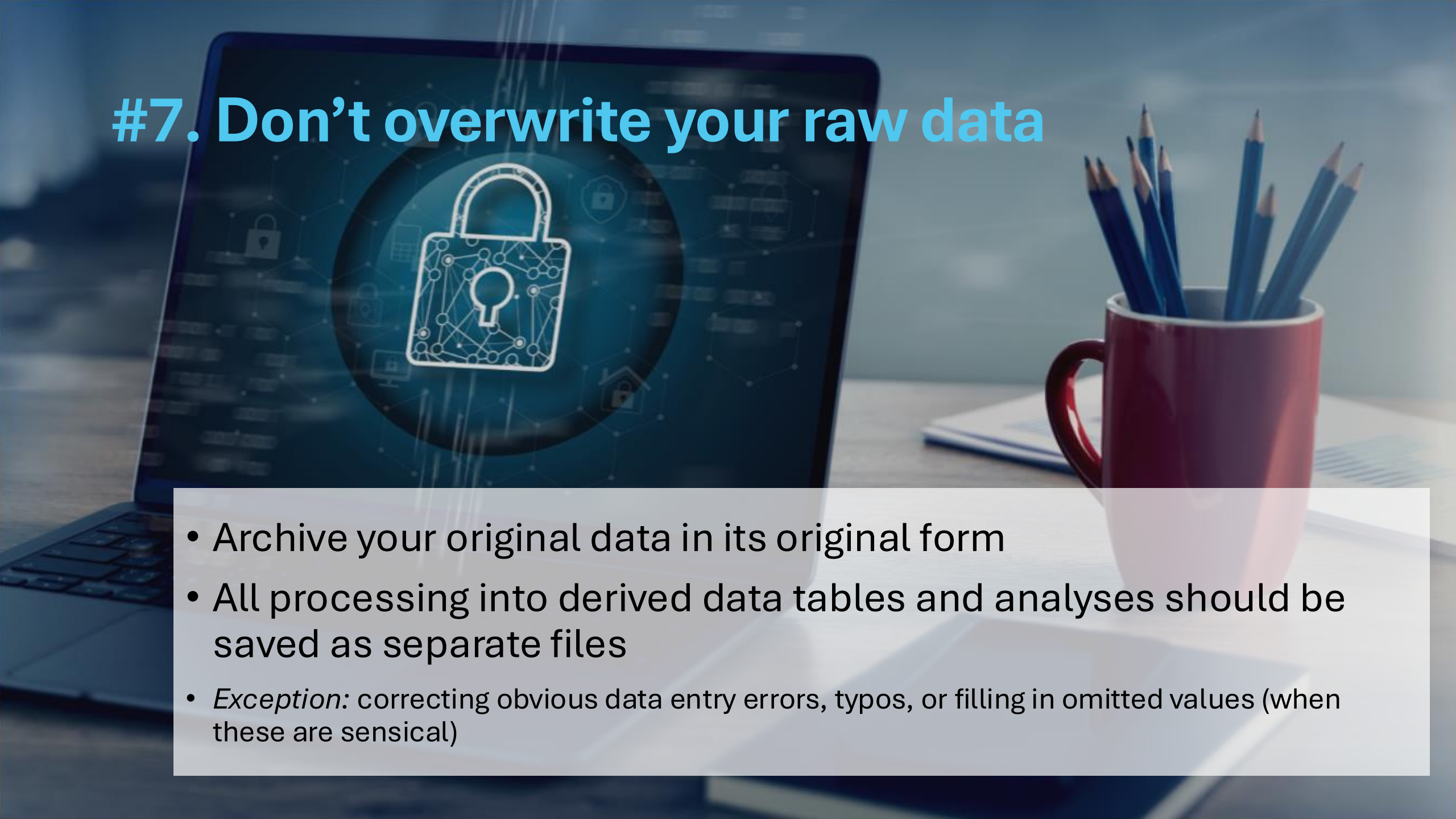


DO NOT go and manually edit your raw data files.

“Data cleaning” is something computers are pretty good at, and processing your data computationally will reduce human error.

- R::tidyverse, R::janitor, pyjanitor

#7. Don’t overwrite your raw data



- Archive your original data in its original form
- All processing into derived data tables and analyses should be saved as separate files
- *Exception:* correcting obvious data entry errors, typos, or filling in omitted values (when these are sensical)

Take accountability

#8. Be accountable for all analysis design choices. Align with literature or be ready to defend differences

- This is YOUR work
- The LLM is not smarter than you
- LLMs can be fantastic teaching tools: take advantage!
  - Have the LLM walk you through each line and give you explanations, assumptions and rationale
- Don’t neglect reading; it’s likely your design has some precedent in the peer reviewed literature
  - Reinventing the wheel comes with risks


Katalin Karikó
© Nobel Prize Org



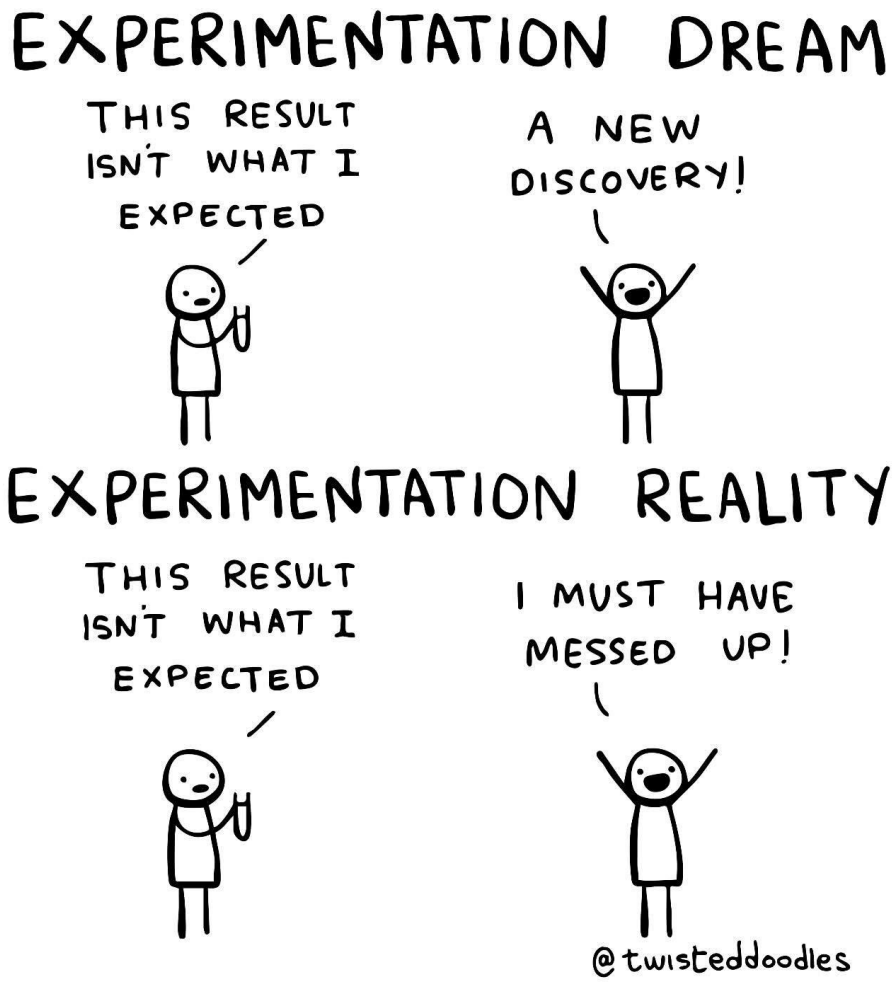
#8. Be accountable for all analysis design choices. Align with literature or be ready to defend differences

Basics you should always be able to answer:

- ❑ What statistical test was used, and why was it appropriate for your data and design?
- ❑ What is the statistical output actually measuring? What is the study interpretation for this?
- ❑ What are the assumptions in your analysis, and have you verified they are met?
- ❑ What does each covariate and parameter in the model represent, and why was it included? If you preprocessed, excluded, or transformed any data, why was this done?
- ❑ (Where applicable) How did you handle data missingness, censoring, outliers, and zeros and what is the biological or study design rationale for doing it the way you did?
- ❑ What is your confidence in the statistical values generated? Are there credible intervals or uncertainty estimates to report?
- ❑ Is the effect size meaningful? Can you describe your results in units you understand?
- ❑ What is the take-home message of any figures you've generated and why/how do they add?

T. Hall © CC-BY 2.0

#9. Sanity check outputs

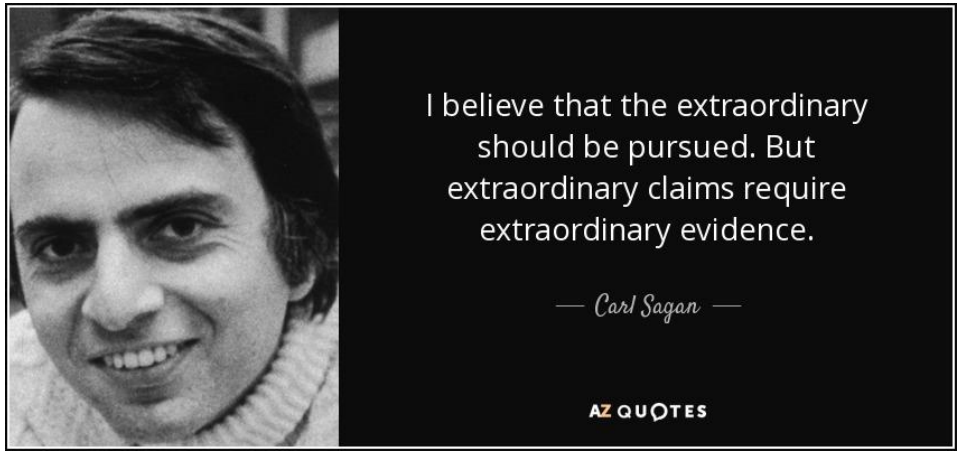


#9. Sanity check outputs

- Your intuition is valuable
  - Biology and medical domains are complex and generate complex data

- “Technically correct” but biomedically impossible or implausible is not correct
  - Look for mishandled or misrepresented data, inappropriate assumptions, or missing uncertainty





#10. Archive a reproducible end-to-end version of your analysis code

- Just ask
- Test it as an end-to-end process and verify all outputs; investigate discrepancies
- Save and date it (**GitHub** and/or as an .Rmd, .qmd, .py, .sh set of scripts)
- Give the clean script back to the LLM and ask for a narrative methods summary
  - this is part of your sanity check/accountability
  - This will form the foundation of your methods section (follow appropriate NIH and external policies for LLM generated text)

10 Simple Rules:

A. Document your design

1. Have a clear hypothesis
2. Describe your design and metadata explicitly
3. Have a validation plan

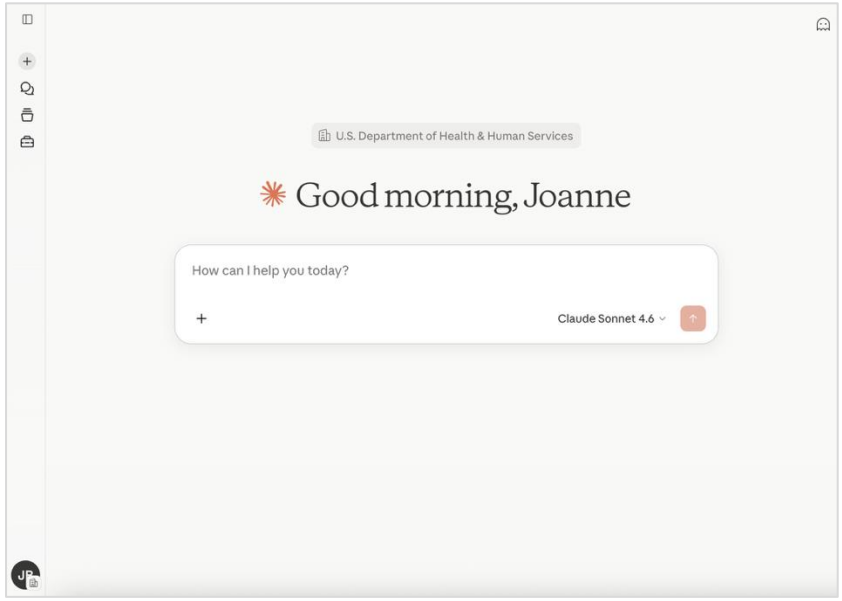
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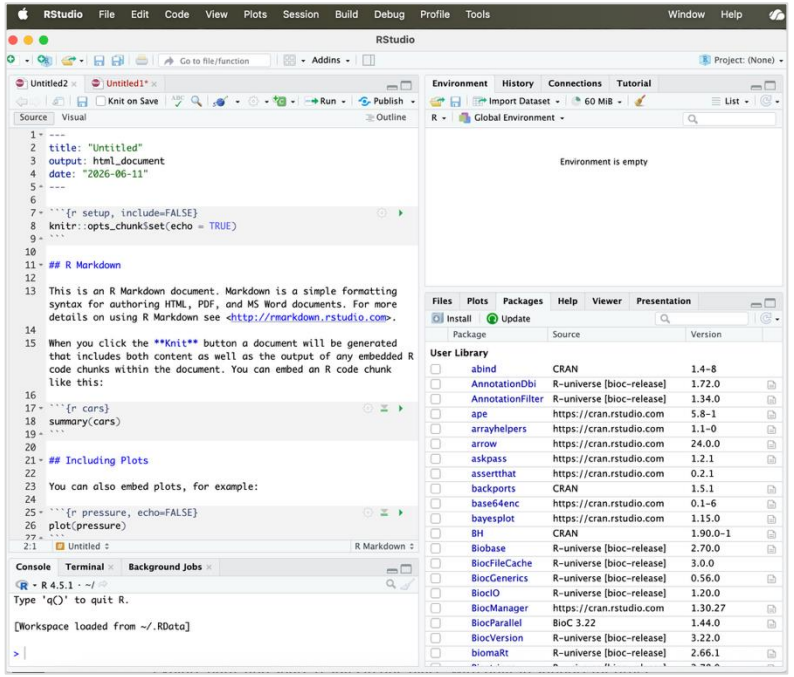
8. Be accountable for all analysis design choices. Align with literature or be ready to defend differences
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Recommendations for step 0

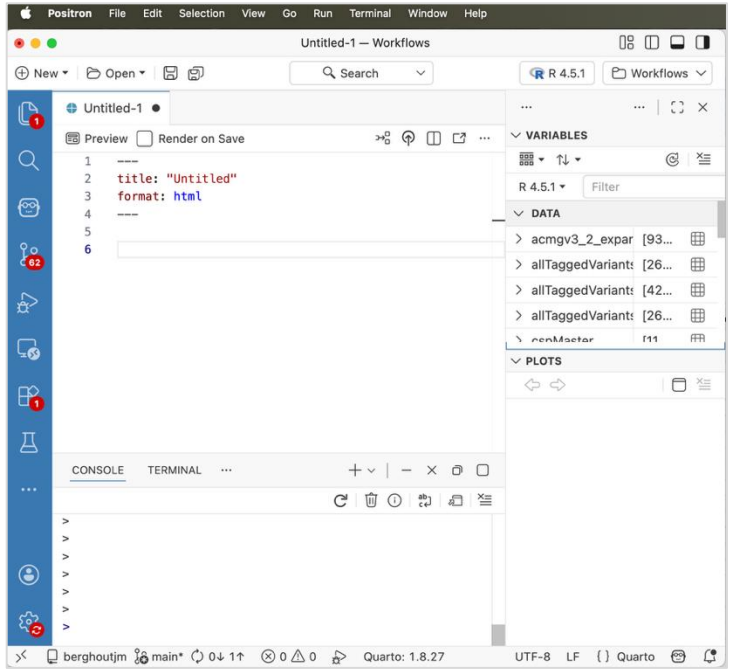


https://claude.hhs.gov/new

You may also be able to get set up with a Github Copilot account, and/or most IDEs (including RStudio and Positron) have integrated AI assistants



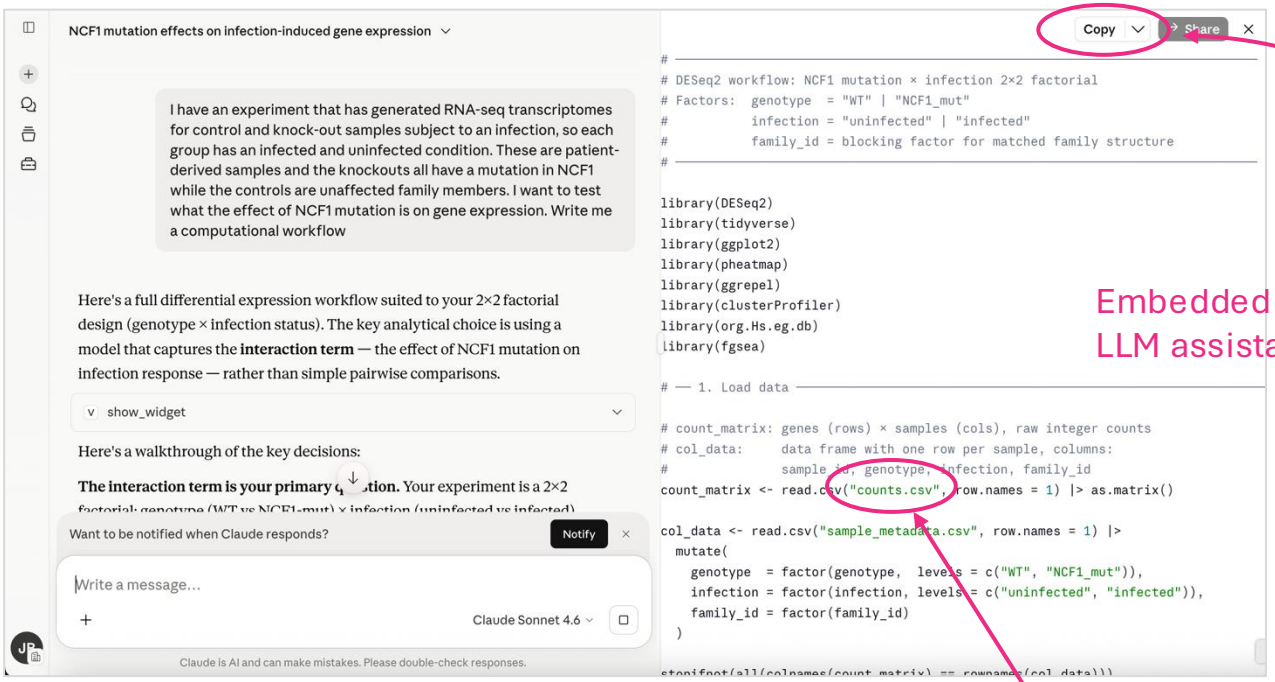
RStudio: available through NIH Self-service



Positron: coming soon!

Next generation of RStudio with better git & AI features

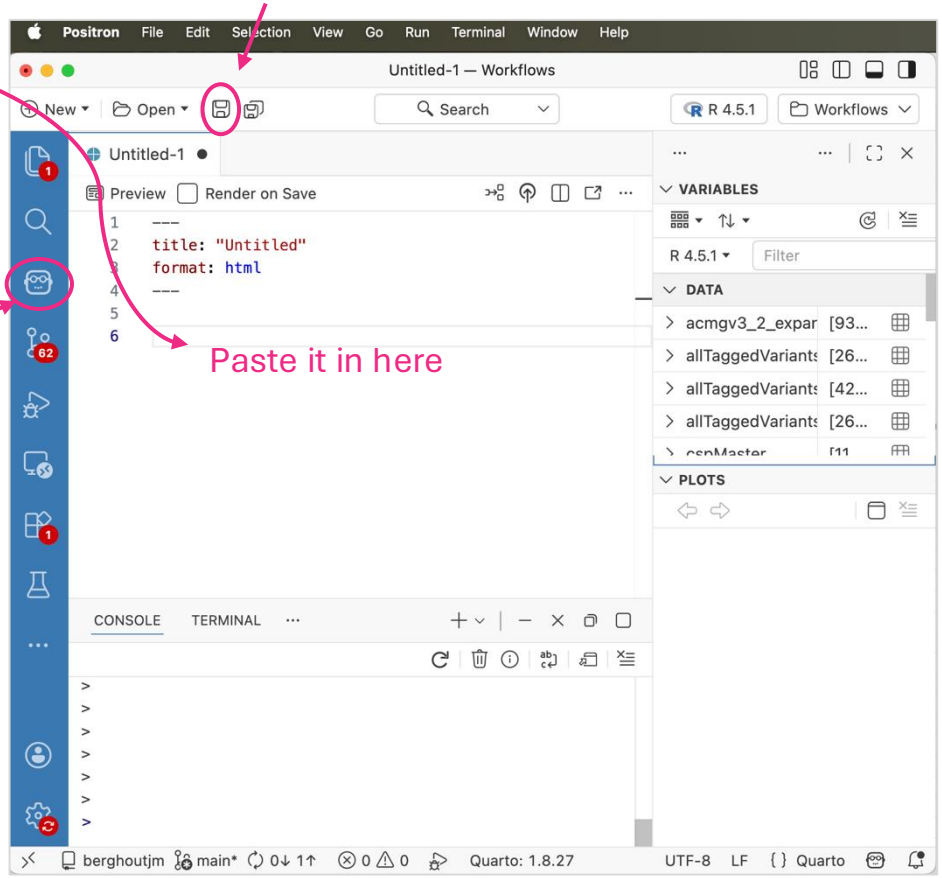
Bringing LLM code to your data is a simple copy-paste



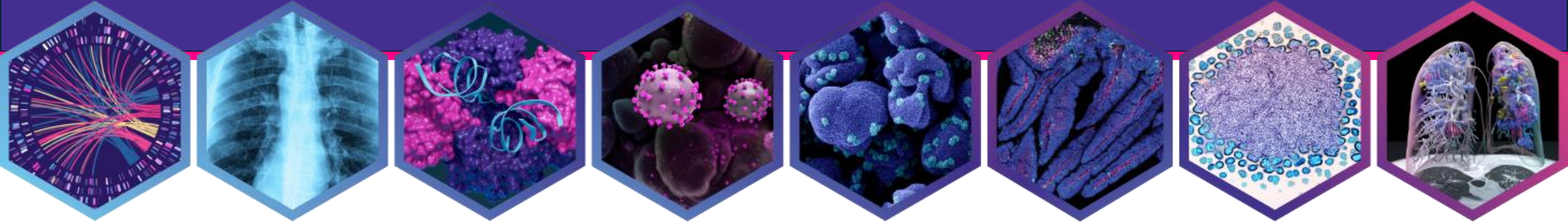
Embedded LLM assistant

Update to the path where your real data is

Save button



Caution! Note that the LLM did *exactly* what I asked and jumped to statistical analysis. It has skipped QC, batch correction, did not ask for covariates, and has not presented me a look at the data. Asking for these steps may need to come from you



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(VPN required)

Upcoming Workshops and Webinars

June 24: Immersive Visualization of Proteins in Virtual Reality with ChimeraX

This interactive workshop introduces participants to the use of virtual reality (VR) in UCSF ChimeraX for immersive visualization of protein structures.

June 25: Statistical Testing in R

This webinar is for biologists, clinicians, and anyone interested in running common statistical analyses on their own data using the R programming language.

Join our ListServ to Hear About Upcoming Trainings

go.nih.gov/6AL6WkH